

GAN-based Image Forgery Detection

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by

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CERTIFICATE

The above-mentioned work entitled “GAN-based Image Forgery Detection” submitted by Soham Pal (23ucc604), Mainak Das (23ucc569) in partial fulfillment of the requirements for a degree of Bachelor of Technology (B. Tech), is hereby certified to be the bonafide work done by the student under my/our supervision and guidance in the Department of Communication and Computer Engineering, The LNM Institute of Information Technology, Jaipur, (Rajasthan) India, during the academic year 2026-2027 and that it has not been submitted anywhere else for getting a similar or higher degree. It is further certified that the standard of the work done is up to the mark.

Date

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Abstract

This project is about a system that uses learning to find parts of images that have been forged or altered. Here we have used a framework called Generative Adversarial Network to achieve the goal,

The Copy-Move Forgery Detection is here implemented on the Comofod dataset. It takes the images from this Comofod dataset and processes them to get masks. The Copy-Move Forgery Detection system make sure that all the images are in proper size, 256×256 pixels. The Copy-Move Forgery Detection system makes a lot of improvements to the images like fixing the colors and shapes. This helps the Copy-Move Forgery Detection system work better and learn from kinds of images and lighting.

Here we have implemented a U-Net Generator. This generator makes high quality masks to show where the fake parts are. Then there is a Patch-Gan Discriminator that checks the masks to make sure they are realistic. The Copy-Move Forgery Detection system learns from a combination of kinds of feedback and losses during traning.Based upon the losses the generator and discriminator updates itself and this continues unless we get a better results.The Copy-Move Forgery Detection system can rebuild the images and how realistic the masks are. The Copy-Move Forgery Detection system also looks at the features of the images to make sure it can find both problems and big structural issues.

The accuracy statistics for the results obtained using the proposed model are Pixel Precision of 0.9048, Pixel Recall of 0.7792, Pixel F1-score of 0.8373, Pixel Accuracy of 0.9821, Pixel IoU of 0.7201, and region mean IoU of 0.6624 on the unseen test data. As seen from the results above, the improvements over baseline models are evident from the better values in all accuracy metrics, implying that the adversarial training and the U-Net framework were effective in detecting forgeries. The qualitative results validate the improved ability to detect forgeries even after processing.

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List of Notations

Mathematical Notation and Symbols

Symbol	Description
<i>Core Variables</i>	
$\mathbf{x}$	Input forged RGB image, $\in \mathbb{R}^{H \times W \times 3}$
$\mathbf{y}, \hat{\mathbf{y}}$	Ground-truth and predicted forgery masks
H, W	Image dimensions (256×256)
p_i	Pixel-wise predicted probability
<i>Model Components</i>	
$G(\cdot)$	Generator (U-Net)
$D(\cdot)$	Discriminator (PatchGAN)
θ_G, θ_D	Trainable parameters of G and D
<i>Loss Functions</i>	
$\mathcal{L}_G, \mathcal{L}_D$	Generator and discriminator loss
$\mathcal{L}_{\text{adv}}$	Adversarial loss component
$\mathcal{L}_{\text{BCE}}$	Binary cross-entropy loss
$\mathcal{L}_{\text{Dice}}$	Dice loss for overlap accuracy
λ	Loss weighting coefficient
<i>Training Parameters</i>	
η	Learning rate (2×10^{-4})
B, E	Batch size and number of epochs
<i>Evaluation Metrics</i>	
P, R, F_1	Precision, Recall, and F1-score
IoU	Intersection over Union
Acc	Pixel accuracy
<i>Other Parameters</i>	
τ	Threshold for binary mask (0.5)
$\sigma(\cdot)$	Sigmoid activation function

Summary of Notation Used in This Report

Chapter 1

Introduction

1.1 The Area of Work

This research is conducted within the junction of the domain of digital image forensics and that of deep learning generation techniques. In particular, the research question concerns the detection and localization of a type of digital image tampering called *copy-move forgery*, which means copying and moving some part(s) of the image in question to hide/remove some other part(s).

Nowadays, the accuracy of digital images has become an important issue for many critical domains. For instance, photographs or video clips used as evidence in court cases or the falsification of one picture in journalism could lead to changing people's view on the whole event. Furthermore, in forensics, no digital images obtained at a crime scene can be forged. Here lies the advantage of using the copy-move forgery – there is no unusual content added to the picture.

Traditional methods of CMFD relied on handcrafted features in terms of SIFT [1], SURF, and DCT-based block descriptors [2]. The similarity between all pairs of the patches/keypoints was computed, and the regions with high similarity were marked as possible forgeries. While this technique was effective in an ideal setting, it proved to be highly vulnerable to any post-processing applied to the copied regions, such as compression, noising, rotation, scaling, and blurring.

Moreover, the recent advances in deep learning methods made a significant contribution to the domain of computer vision in general, and specifically to forensic imaging. Indeed, CNN models learn hierarchical features directly from the input dataset that are best suited for a certain task. Thus, they tend to be less prone to various distortions than engineered feature descriptors. Recently, Generative Adversarial Networks (GANs), [3] where the generator is pitted against the discriminator in a minimax game framework, have shown outstanding performance in tasks requiring image-to-image translation. There is a specific class of GANs,

known under the name of Pix2Pix [4], that can be readily applied to forgeries detection problems. In Pix2Pix GANs, the generator generates an accurate spatial mask of the forgery given an image, while the PatchGAN discriminator ensures the local spatial realism of the mask.

The developments in the field of deep learning in the field of computer vision have been very crucial in changing the approach towards image forensics. The key point that makes convolution neural networks more efficient compared to handcrafted feature extraction algorithms is their capability to learn hierarchically organized features that are specific to the problem under consideration. Another architecture that involves two modules known as a generator and a discriminator playing a game of minimizing and maximizing some function respectively and called Generative Adversarial Networks (GANs) [3] have been shown to be highly efficient in solving problems related to generating new images from existing images. One class of GANs is known as Pix2Pix GAN where the generator learns a pixel-wise mapping from the input to the forged image while the PatchGAN discriminator takes care of the local realism.

1.2 Problem Addressed

Image manipulations are easily possible with the help of software applications that are freely available. Of all the different types of image manipulations such as collage, inpainting, contrast alteration, removal of objects, etc., copy and move becomes particularly problematic because of the simple reason that there is no addition of any extra information in the image through this process.

The main issue that the proposed project aims at addressing is that of *discriminative spatial localization*. It means that for any given fake image, one would have to develop a binary mask depicting all the areas that have been manipulated through copying/moving processes. This particular aspect is much harder than simple binary classification tasks related to images.

Several specific sub-problems arise within this broader challenge:

- **Class imbalance:** For almost all the samples in this dataset, the forgery area is much smaller compared to the entire size of the image. Training the classifier by simply employing naive cross entropy loss will cause the network to predict the background class and get it right with ease.
- **Post processing robustness:** Images from the CoMoFoD dataset have been subjected to post processing techniques like JPEG compression at varying quality levels, adding of Gaussian noise, adjusting contrast and introducing color jittering.
- **Transformation geometry:** The copy part need not essentially be a translated version of the original part. The copy can be transformed in many other ways like rotation and reflection. Therefore, performing nearest neighbor matching becomes infeasible.

These problems can be addressed by our method which incorporates appropriate architectural decisions (U-net with skip connections for recovering spatial information), an appropriately selected multi-loss function (BCE, Dice Loss, adversarial loss), and careful evaluations based on both pixel-wise and region-based metrics.

1.3 Existing Systems

Copy-move forgery detection methods can be broadly categorised into three generations: classical feature-based methods, supervised CNN-based detectors, and GAN-based approaches.

1.3.1 Classical Feature-Based Methods

Earlier work in CMFD relied on the knowledge that a copy-move attack creates a new copy of a region somewhere else in the same image. In block based detection schemes, regions are divided into overlapping blocks. Each of these blocks is assigned a feature vector (DCT transform coefficients or PCA-based patches) which are sorted in a lexicographic manner and then compared to detect forgeries [2]. The method proposed by [1] showed how DCT transform blocks can be matched efficiently using the idea of sorting; however, it performed poorly under various geometrical and photometric attacks.

Keypoint-based algorithms were first developed by the SIFT descriptors [5], improving robustness against rotations and scaling by detecting and matching keypoints rather than matching patches densely but sparsely. [6] proposed a technique known as PatchMatch-based dense matching, which bridges the gap between keypoint-based and dense matching techniques. This algorithm offers dense matching capabilities but can withstand moderate levels of affine distortion. Nonetheless, despite these improvements, the conventional approach is limited due to its inability to represent complex spatial relationships and susceptibility to heavy JPEG compression.

1.3.2 CNN-Based Detection

The model of DV-Net which detects inconsistency regions between the fake image and reconstructed image without referencing other images using a dual-view network for joint reasoning was introduced by [7]. The approach called SSCMFDNet for detecting inconsistencies between image patches using self-correlation maps without requiring explicit patch matching was developed by [8]. The technique of deep feature extraction in combination with traditional matching was employed in the work by [9].

The methods based on CNNs for detecting manipulated images usually formulate the task as a binary segmentation problem where the decoder generates forgery masks from features

extracted from the image by the encoder. While outperforming traditional methods on common datasets, such methods face challenges with boundary localization owing to pooling operations used in encoders.

1.3.3 GAN-Based and Hybrid Approaches

Some recent works have investigated GANs in forgery localization. [10] proposed a novel Dual Homology-Aware GAN (DH-GAN) that employs structural consistency losses between homologous locations to improve mask prediction performance. Moreover, [11] proposed Dual Stream-Dual Branch GAN, which leverages two branches in the frequency and spatial domains and fuses evidence via adaptive gates. Finally, [12] proposed Multi-Granularity Feature Consistency Detection Network (MGCFDN), which works at multiple spatial granularities for detecting large and small forged patches.

The work of [13] demonstrated that combining adversarial learning and leveraging the U-Net backbone could significantly boost the performance of the latter technique, confirming our experimental findings. This GAN discriminator acts as a learnable loss function, allowing the model to learn high-order structure that cannot be captured by pixel-level losses. In this regard, our technique mainly focuses on the latter category. Our architecture follows the Pix2Pix-based GAN framework [3] where the generator architecture includes the U-Net with residual connections and a PatchGAN discriminator to supervise the pixels.

1.4 Literature Context

The subject of image forgery localization has been studied thoroughly throughout many decades. The field started from the frequency analysis of block-level information and went further on to detecting keypoints, followed by learning segmentation using CNNs, and lately, localization using GANs. Nevertheless, there is always one thing in common in all the aforementioned literature that no technique could be considered as the optimal choice for all kinds of forgery attacks, transformations, and data sets. While block-based techniques are effective for attacks that involve translations only and do not require post-processing, they will be incapable of handling attacks with rotations or compressions. In addition, while the keypoint method is capable of dealing with different geometric transformations, it generates highly sparse masks, thus making it impossible to detect regions. Contrarily, while the output generated using CNNs is dense enough, it lacks spatial sharpness necessary for detecting regions.

Adding the task of adversarial training as an auxiliary objective solves this issue because it ensures that the generator needs to generate a binary mask which is indistinguishable from the real one locally, thus forcing it to produce sharper boundaries.

Chapter 2

Literature Review

2.1 Introduction

The detection of image tampering from the problem of obscure digital signal processing is now a relevant issue due to social, legal, and journalism aspects. Considering the availability of editing software and abundance of social media, a forged image may reach millions of people until its verification. Among all kinds of manipulations on images, the copy-move forgery belongs to the ones which can cause the most harm as this kind of manipulation does not contain anything foreign to the picture; rather, the content is taken from elsewhere in the same picture.

In this chapter, we give an overview of the current level of knowledge about CMFD techniques. This chapter focuses on the literature review concerning the CMFD area, which consists of the following sections: (1) classic approaches based on features, (2) benchmark datasets, (3) segmentation based on convolutional neural networks, (4) encoder-decoder models, (5) localization using generative adversarial networks, and (6) evaluation methods. Thirty references are given in this chapter, including pioneering studies and current literature.

2.2 Foundational and Classical Copy-Move Forgery Detection

2.2.1 The Block-Matching Paradigm

Detection of duplicated areas within images was originally studied by [14]. They developed the first block-matching algorithm for CMFD that partitions an image into overlapping square blocks encoded with compact descriptors based on the DCT and reports pairs of blocks with very similar descriptors. Although this approach pioneered the area, it was computationally too costly and did not allow for any image processing during the duplication stage.

[15] made a next step toward solving the problem by applying PCA for decreasing dimensionality of descriptors. The results obtained from synthetic images were used for building the first set of metrics.

First of all, as mentioned in [1], the idea of early exit was applied to the procedure of lexicographic sorting, leading to a reduction of computation costs. However, according to the results of the comprehensive benchmark study performed in [2] on five DCT block descriptors and two methods based on keypoints, it can be seen that, despite their effectiveness, the problem of poor robustness remains relevant in regard to block descriptors when rotation, scaling, and JPEG compression are applied. This paper is considered the benchmark paper for evaluating baseline CMFD methods and showed the need for developing more robust representations.

[16] proposed to use texture descriptors as an alternative to DCT descriptors due to their higher tolerance to small changes in colors and illuminations. Nonetheless, similar to block descriptors, texture descriptors do not possess any properties of geometric transformation invariance.

2.2.2 Keypoint-Based Approaches

The issue of geometrical sensitivity was solved through the adoption of keypoint techniques due to their ability to provide scale-invariant and rotation-invariant sparse features. [5] present a survey of CMFD solutions based on keypoint techniques such as SIFT, SURF, and ORB which show high invariance to affine transformations but provide just sparse maps, unable to distinguish the edges of forgeries. In total, they tested 40 algorithms, finding out that none of them could become superior under all types of evaluation.

According to [17], the combination of PCET descriptors with target keypoints provided higher resistance to rotation and reflection forgeries.

In [18], the authors performed a comprehensive review of CMFD problems and potential research directions, determining the open problems in this area to include: geometric transformation resilience, JPEG compression sensitivity, computational efficiency, and moving from binary detection to exact spatial location. These problems provide the motivation for applying deep learning techniques to solve the problem end-to-end.

2.2.3 PatchMatch and Dense Correspondences

In [6], the authors proposed a dense-field CMFD approach inspired by the PatchMatch [19] method, which combines the completeness of block-wise CMFD approaches and the efficiency of approximate nearest neighbour search. Compared to keypoint-based CMFD methods, the proposed dense correspondence field better covers large forged regions. However, PatchMatch does not work well for small forged regions and is sensitive to the patch size setting.

2.3 Benchmark Datasets for CMFD Evaluation

Experimental results from CMFD studies highly depend on the selected evaluation data set. There are three sets which have been widely accepted as standard test sets:

MICC-F220 and MICC-F2000 [2]: MICC-F220 includes 220 images with forgery operations restricted to translation only, without any post-processing. Similarly, MICC-F2000 contains 2000 images with only translation-based forgeries without post-processing. Both data sets can be used to measure the basic detection performance but do not represent real-life tampering scenarios.

CoMoFoD [20]: Copy-Move Forgery Database contains 260 pairs of forged and pristine images together with their binary masks as ground-truth. The novelty of the data set is that for each pair, there are five different types of post-processes applied, which are (i) Gaussian noise and JPEG compression (quality = 70-95), (ii) geometric manipulation (rotation, scaling, mirroring), (iii) contrast and brightness adjustment, (iv) color space reduction, and (v) combinations of above manipulations. Therefore, CoMoFoD is the most difficult and realistic CMFD test set, and we will use.

COVERAGE [7]: This 100-image dataset was created to test forgery-detection algorithms based on self-similarity with the inclusion of *distractor regions*, which are image patches that are visually similar to the forged patches but not duplicates. This dataset is often used as an out-of-distribution data set when training on the CoMoFoD dataset.

CASIA [21]: The CASIA dataset includes forgery types like copy-move and splicing, which makes it an appropriate benchmark for multi-forgery types.

2.4 Deep CNN-Based Detection

2.4.1 CNN Feature Representations

The use of CNNs in forgery detection involved a change in paradigm from handcrafted features to learned features. [9] reported a successful hybrid approach using CNNs combined with traditional matching, although such an approach has the disadvantages of being a two-stage process and needing large amounts of labeled data to train the CNN.

In their survey of CMFD methods based on deep learning from 2015-2022, [22] found U-Net and attention-based models to be the most promising research avenues. Their analysis indicates that fully trainable approaches significantly outperform the hybrid feature-engineering approaches.

2.4.2 Self-Correlation and Attention Mechanisms

[8] designed the SSCMFDNet, which consists of a self-correlation layer that performs comparison of similarity of features at all the pairs of spatial locations and indirectly performs copy-move matching at the feature level. This innovation in network architecture brings the solution to its roots in terms of copy move forgery detection: detection of duplications. Self-correlation is especially helpful for forgeries that involve geometrical transformation since the feature-level matching is more invariant to transformations compared to the pixel-level one.

[23] proposed BusterNet, which was the first architecture to find both the *source* and *target* regions simultaneously. BusterNet used two branches for different tasks: manipulation detection and similarity search branches. BusterNet was trained on a large dataset synthesized specially for this task.

[24] examined self-supervised pre-training for learning copy-move forgery detection features based on a prediction task for determining whether the given image patches are duplicates of one another. It alleviates the dependency on the labeled forgery datasets and enhances the generalization capability on unseen forgery types.

2.4.3 Dual-View and Multi-Branch CNNs

DV-Net [7] is a dual-view structure that takes the input image into consideration through the direct appearance and residual branches in parallel. The first one detects low-level features from the RGB channels of the image, while the second one increases the high-frequency manipulation signatures. They are integrated at several levels, generating a dense forgery map. Although it performs well on COVERAGE without adversarial training, it cannot generate crisp boundaries.

2.5 Encoder-Decoder Architectures for Dense Prediction

Since [25] proposed the U-Net architecture for biomedical image segmentation, it has emerged as the standard backbone network for dense prediction problems in forensics. The U-Net architecture is specifically designed to overcome the trade-off between global information perception and local positioning through its encoder-decoder architecture with skip connections.

[26] utilized the U-Net architecture for detecting tampered images and found that the skip connections make it possible to precisely locate the boundaries of forged regions even in high-resolution images. They proposed RRU-Net, which used ringed residual connections that help stabilize gradient flow in skip connections.

The use of region-level contrastive learning was put forth by [27] in the context of semisupervised segmentation, indicating the benefit of imposing consistency between predictions from

various augmentations of the same region. Although used in the case of natural image segmentation, this concept of region-level consistency is consistent with the CoMoFoD benchmark that evaluates region-level IoU.

The basic work on residual learning by [28] included skip connections in classification networks, which have been implemented in U-Net encoder blocks. Residual learning has been effective in other tasks too, including forgery detection. In forgery detection, the generator has to learn to detect anomalies in regions that mostly conform to authentic images.

2.6 Generative Adversarial Networks for Forgery Localisation

2.6.1 The GAN Framework

The Generative Adversarial Network [3], introduced by Goodfellow et al., formalises learning as a two-player minimax game between a generator G that produces synthetic data and a discriminator D that distinguishes real from synthetic. The adversarial objective is:

$$\mathcal{L}_{\text{cGAN}}(G, D) = \mathbb{E}_{x,y}[\log D(x, y)] + \mathbb{E}_x[\log(1 - D(x, G(x)))] \quad (2.1)$$

The conditional GAN formulation [29] conditions both G and D on an additional input, enabling image-to-image translation tasks. The Pix2Pix framework [4] operationalises cGANs for paired image translation with a U-Net generator and a PatchGAN discriminator, achieving state-of-the-art results across diverse image translation benchmarks. This framework maps directly to the CMFD problem, where the input is a forged image and the target output is its forgery mask.

2.6.2 PatchGAN Discriminator

The PatchGAN discriminator [4] uses local image patches for classification instead of considering the whole image, thereby creating a real/fake spatial map of local decisions where each cell represents a 70×70 pixel region. The average of these local decisions provides the discriminator loss. This model is highly suitable for forgery localization tasks since the border between the forged and genuine parts is inherently a local concept.

2.6.3 Modern GAN-Based CMFD Architectures

DH-GAN, or Dual Homology-Aware GAN, was suggested in [10]. This architecture is capable of processing input feature maps in both the spatial and the frequency domain. Specifically, a homography layer is added to encode spatial homologies among encoder feature maps due

to the presence of self-similarity in copy-move forgeries. Despite superior performance, dual-domain GANs need more computational resources than single-domain architectures.

The Dual-Stream Dual-Branch GAN was introduced in [11]. The authors proposed to process input images separately via two branches for finding anomalies related to texture and those related to semantics. An adaptive attention mechanism allows fusing the two branches before predicting the mask of forgeries, whereas an auxiliary feature matching loss is used in addition to the standard GAN loss function.

The Multi-Granularity Feature Consistency Detection Network (MGCFDN) [12] has been proposed by utilizing the forgery detector in multiple granularities and integrating the outputs of masks at all granularities. This design allows MGCFDN to detect large-scale and small-scale forgery areas, which can be considered a critical weakness in the existing single-granularity architecture.

Combining the U-net with adversarial training has shown remarkable performance [13]. They reported an improvement in IoU and F1 scores compared to using either technique separately, especially on highly compressed JPEG forgery images.

[30] provided a comprehensive review of deep-learning-based CMFD methods from 2018 to 2023, concluding that GAN-based methods consistently outperform pure discriminative CNN approaches, attributing this to the adversarial loss acting as a learned, high-order structural regularizer that pixel-wise losses cannot replicate.

2.7 Evaluation Methodology

2.7.1 Pixel-Level Metrics

The standard protocol for CMFD evaluation uses binary segmentation metrics derived from the confusion matrix [2]:

Precision measures the fraction of detected forgery pixels that are genuinely forged:

$$P = \frac{TP}{TP + FP} \quad (2.2)$$

Recall measures the fraction of genuinely forged pixels that are detected:

$$R = \frac{TP}{TP + FN} \quad (2.3)$$

F1-Score is the harmonic mean of Precision and Recall:

$$F_1 = \frac{2 \cdot P \cdot R}{P + R} \quad (2.4)$$

Pixel Accuracy is the fraction of all pixels correctly classified:

$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.5)$$

IoU (Jaccard Index) directly measures the overlap between predicted and ground truth masks:

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (2.6)$$

2.7.2 Region-Level Metrics

All pixels are considered equal in pixel-based metrics, which may hinder the model from detecting distinct forging cases. In contrast, the Region mean Intersection over Union (Region-mIoU), utilized in [23] and incorporated into our evaluation paradigm, calculates IoU on a per-ground-truth component basis via greedy best-matching to the predicted component, followed by averaging the result across all images.

2.8 Summary and Positioning of Proposed Work

Table 2.1 presents a summary of the 30 papers reviewed in this chapter, along with their corresponding methodology, dataset used for evaluation, and major limitations. As such, the proposed GAN-CMFD algorithm takes its rightful place within this design landscape, borrowing the architecture and approach demonstrated as viable in [13] and [4], implementing an innovative multi-component loss function to deal with class imbalance, and testing against the CoMoFoD dataset with a leak-free base image group split. In contrast to DH-GAN [10], MGCFDN [12], and dual-branch GAN [11], the GAN-CMFD method does not include frequency domain analysis and multiscale fusion.

Reference	Approach	Dataset	Limitation
Fridrich et al. [14]	DCT blocks	Custom	No post-processing
Popescu [15]	PCA blocks	Custom	High FP rate
Lin et al. [1]	DCT sorting	Custom	No geometric robustness
Christlein [2]	Multi-descriptor	MICC	Benchmark only
Ardizzone [16]	Texture	Custom	JPEG sensitive
Teerakanok [5]	SIFT survey	Multiple	Sparse masks
Zhang [17]	PCET	CoMoFoD	Parameter tuning
Warif [18]	Survey	Multiple	Survey only
Cozzolino [6]	PatchMatch	MICC	Small region issue
Zhu (CoMoFoD) [20]	Dataset	CoMoFoD	Benchmark only
Dong (CASIA) [21]	Dataset	CASIA	Mixed forgeries
Zare [9]	Hybrid CNN	CoMoFoD	Complex pipeline
Nguyen [8]	SSCMFDNet	CoMoFoD	High memory
Wu [23]	BusterNet	CASIA	Needs large data
Zhao [24]	Self-supervised	CoMoFoD	Pretraining cost
Weng [7]	DV-Net	COVERAGE	No adversarial training
Raju [22]	Survey	Multiple	Survey only
Kashyap [30]	Survey	Multiple	Survey only
Bi [26]	RRU-Net	CASIA	Not CMF-specific
He [28]	ResNet	ImageNet	Generic backbone
Huang [27]	Contrastive	Custom	Needs unlabeled data
Ronneberger [25]	U-Net	Medical	Domain shift
Goodfellow [3]	GAN	Synthetic	No spatial cond.
Isola [4]	Pix2Pix	Multiple	General task
Zhu (CycleGAN) [29]	CycleGAN	Multiple	Unpaired setup
Alfadli [13]	U-Net+GAN	CoMoFoD	Instability
Liu [10]	DH-GAN	CoMoFoD	Slow training
Bai [11]	Dual GAN	Multiple	Data heavy
Tu [12]	MGCFDN	CoMoFoD	High latency
Proposed	U-Net+PatchGAN	CoMoFoD	Single-scale

TABLE 2.1: Comparative Summary of CMFD Methods

Chapter 3

Proposed Work

3.1 Overview of the Proposed Solution

The proposed system depicts a copy-move forgery detection as a conditional image-to-image translation problem. Given a forged input image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, the goal is to predict a binary segmentation mask $\hat{\mathbf{y}} \in \{0, 1\}^{H \times W}$ where pixel value 1 indicates a manipulated (forged) region and 0 indicates an authentic region.

The overall pipeline which is followed in this project is illustrated in Figure 3.1. At a high level, the system consists of three major components:

1. **Data Preprocessing and Pipeline:** The CoMoFoD dataset, including the input images and their corresponding mask images, is fed into the system, resampled to 256×256 pixels, and then split into training, validation, and testing datasets based on the group of base images to avoid data leakage.
2. **Generator (U-Net):** A deep neural network architecture known as the U-Net model, with skip connections, takes the forged image as an input and produces a one-channel logit map, from which the forgery mask can be extracted by applying a sigmoid threshold function.
3. **Discriminator (PatchGAN):** This is a full convolutional network that gives the generator feedback via adversarial gradients.

The system is trained end-to-end using a composite loss function combining adversarial supervision, segmentation accuracy, and region overlap objectives.

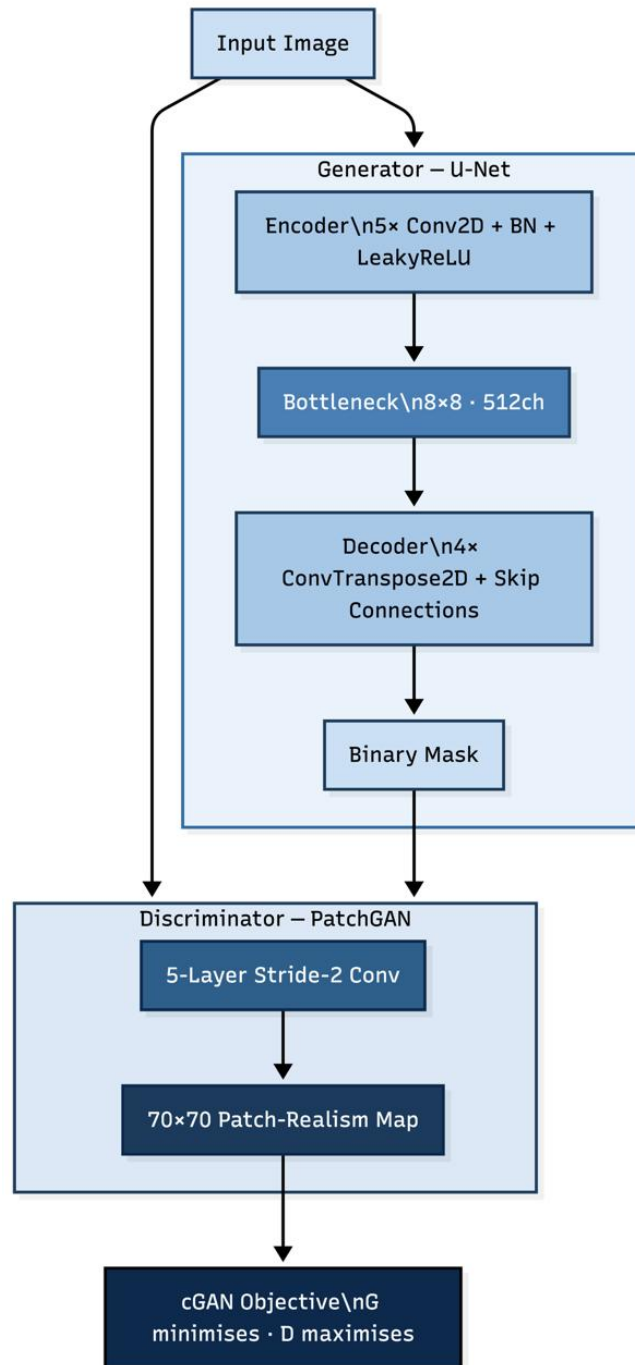


FIGURE 3.1: Proposed GAN-CMFD architecture: U-Net generator with PatchGAN discriminator for binary forgery mask prediction.

3.2 Dataset and Preprocessing Pipeline

3.2.1 CoMoFoD Dataset

CoMoFoD (Copy-Move Forgery Database) [20] dataset is used as the benchmark dataset throughout this thesis. The dataset has 260 groups of base images that consist of clean original image(s), corresponding forgeries, and exact binary masks that denote regions where copy-move forgery occurs. There are five different types of forgery in CoMoFoD. The forgery includes (i) Gaussian noise addition and JPEG compression, (ii) geometrical manipulation (rotation, scaling, mirroring), (iii) contrast and brightness adjustment, (iv) reduction in color space dimensionality, and (v) combination of those.

We employ CoMoFoD_small_v2, which contains appropriate size of images to train our networks. The following figure 3.2 shows data splits of training, validation, and test datasets.

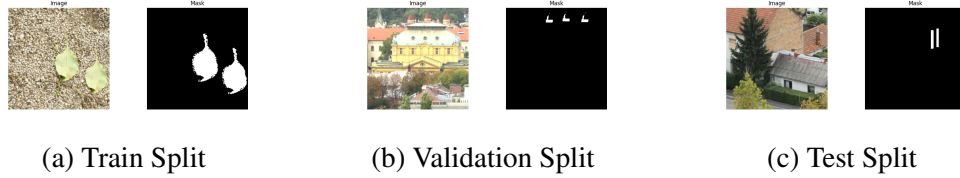


FIGURE 3.2: Dataset loader visualisations showing image-mask pairs from the three splits of CoMoFoD-small.

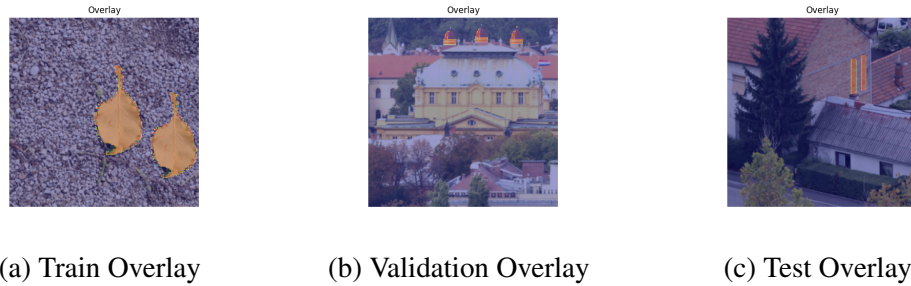


FIGURE 3.3: Overlay visualisations showing forgery mask superimposed on image for each dataset split.

3.2.2 Data Splitting Strategy

The primary challenge in splitting CMFD datasets is *data leakage*: multiple post-processing variants of the same base image, if distributed across train and test splits, allow the network to effectively “memorise” images rather than generalise. To prevent this, we use a **base-image-grouped split**: all variants of the same base image are assigned to the same split.

$$\text{Split ratios: Train} = 70\% \mid \text{Val} = 15\% \mid \text{Test} = 15\% \quad (3.1)$$

A fixed random seed (`random.seed(42)`) ensures reproducibility. Algorithm 1 details the dataset loading and splitting procedure.

Algorithm 1 CMFDDataset: Data Loading and Split Construction

Require: `image_dir`, `mask_dir`, `resize = 256`

```

1: image_list  $\leftarrow \emptyset$ 
2: for each file f in SORTED(image_dir) do
3:   if f has valid extension and ``_F`` in f then
4:     base  $\leftarrow f.$ SPLIT(```_F``)[0]
5:     mask_path  $\leftarrow$  mask_dir/(base + ``_M`` + ext)
6:     if mask_path exists then
7:       image_list.APPEND(f)
8:     end if
9:   end if
10: end for
11: base_ids  $\leftarrow \{f.$ SPLIT(```_F``)[0] : f  $\in$  image_list $\}$ 
12: RANDOM.SHUFFLE(base_ids, seed=42)
13: n  $\leftarrow |base\_ids|$ 
14: train_ids  $\leftarrow base\_ids[: 0.7n]$ 
15: val_ids  $\leftarrow base\_ids[0.7n : 0.85n]$ 
16: test_ids  $\leftarrow base\_ids[0.85n :]$ 
17: return (train_ids, val_ids, test_ids)

```

3.2.3 Image Preprocessing

Algorithm 2 describes the per-sample preprocessing applied consistently to all splits.

Algorithm 2 Image and Mask Preprocessing

Require: `image_path`, `mask_path`, $s = 256$, $\tau = 0.5$

```

1: I  $\leftarrow$  PIL.OPEN(image_path).CONVERT("RGB")
2: I  $\leftarrow$  RESIZE( $s \times s$ , bilinear)(I)
3: x  $\leftarrow$  I.ToTensor()  $\in [0, 1]^{3 \times s \times s}$ 
4: M  $\leftarrow$  PIL.OPEN(mask_path).CONVERT("L")
5: M  $\leftarrow$  RESIZE( $s \times s$ , nearest)(M)
6: mraw  $\leftarrow$  M.ToTensor()  $\in [0, 1]^{1 \times s \times s}$ 
7: y  $\leftarrow$  (mraw >  $\tau$ ).FLOAT()  $\in \{0, 1\}^{1 \times s \times s}$ 
8: return (x, y)

```

No data augmentation is applied offline, since CoMoFoD’s multiple post-processing variants per base image inherently provide augmentation-like diversity. Future iterations will consider online augmentation.

3.3 Generator Architecture: U-Net

3.3.1 Overall Structure

The generator is a U-Net architecture consisting of an encoder and decoder with skip connections. The encoder down-samples the image, while the number of channels increases with the same process continuing until it reaches the bottleneck layer. The decoder performs up-sampling through deconvolution operations where, at each level, it is enhanced by skip connections from the respective encoder layer.

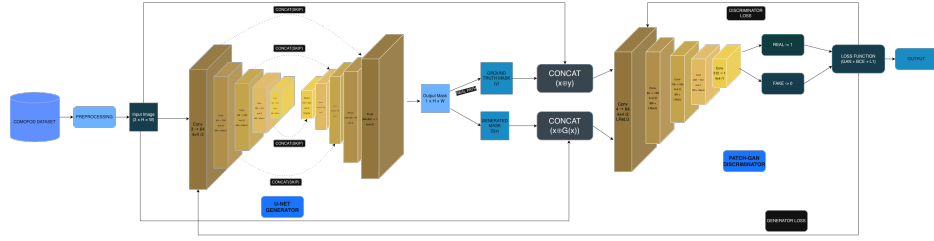


FIGURE 3.4: Detailed U-Net Generator Architecture: encoder (left), bottleneck (centre), and decoder with skip connections (right). Each box shows the spatial dimension and number of channels at that stage.

Algorithm 3 U-Net Generator: Forward Pass

Require: Input image $\mathbf{x} \in \mathbb{R}^{B \times 3 \times 256 \times 256}$	$\triangleright$ Encoder (Downsampling Path)
1: $\mathbf{d}_1 \leftarrow \text{UNETBLOCK}_{\text{down}}(\mathbf{x}, 3 \rightarrow 64, \text{no BN})$	$\triangleright 128 \times 128$
2: $\mathbf{d}_2 \leftarrow \text{UNETBLOCK}_{\text{down}}(\mathbf{d}_1, 64 \rightarrow 128)$	$\triangleright 64 \times 64$
3: $\mathbf{d}_3 \leftarrow \text{UNETBLOCK}_{\text{down}}(\mathbf{d}_2, 128 \rightarrow 256)$	$\triangleright 32 \times 32$
4: $\mathbf{d}_4 \leftarrow \text{UNETBLOCK}_{\text{down}}(\mathbf{d}_3, 256 \rightarrow 512)$	$\triangleright 16 \times 16$
5: $\mathbf{d}_5 \leftarrow \text{UNETBLOCK}_{\text{down}}(\mathbf{d}_4, 512 \rightarrow 512)$	$\triangleright 8 \times 8$ bottleneck
	$\triangleright$ Decoder (Upsampling Path with Skip Connections)
6: $\mathbf{u}_1 \leftarrow \text{UNETBLOCK}_{\text{up}}(\mathbf{d}_5, 512 \rightarrow 512)$	$\triangleright 16 \times 16$
7: $\mathbf{u}_1 \leftarrow \text{CONCAT}([\mathbf{u}_1, \mathbf{d}_4], \text{dim} = 1)$	$\triangleright 1024$ channels
8: $\mathbf{u}_2 \leftarrow \text{UNETBLOCK}_{\text{up}}(\mathbf{u}_1, 1024 \rightarrow 256)$	$\triangleright 32 \times 32$
9: $\mathbf{u}_2 \leftarrow \text{CONCAT}([\mathbf{u}_2, \mathbf{d}_3], \text{dim} = 1)$	$\triangleright 512$ channels
10: $\mathbf{u}_3 \leftarrow \text{UNETBLOCK}_{\text{up}}(\mathbf{u}_2, 512 \rightarrow 128)$	$\triangleright 64 \times 64$
11: $\mathbf{u}_3 \leftarrow \text{CONCAT}([\mathbf{u}_3, \mathbf{d}_2], \text{dim} = 1)$	$\triangleright 256$ channels
12: $\mathbf{u}_4 \leftarrow \text{UNETBLOCK}_{\text{up}}(\mathbf{u}_3, 256 \rightarrow 64)$	$\triangleright 128 \times 128$
13: $\mathbf{u}_4 \leftarrow \text{CONCAT}([\mathbf{u}_4, \mathbf{d}_1], \text{dim} = 1)$	$\triangleright 128$ channels
14: $\hat{\mathbf{z}} \leftarrow \text{CONVTRANSPOSE2D}(\mathbf{u}_4, 128 \rightarrow 1)$	$\triangleright 256 \times 256$ logits
15: return $\hat{\mathbf{z}}$	$\triangleright$ Raw logits; sigmoid applied at inference

3.3.2 UNetBlock Component

Algorithm 4 UNetBlock: Encoder and Decoder Building Block

Require: in_channels c_{in} , out_channels c_{out} , direction $\in \{\text{down}, \text{up}\}$, use_bn $\in \{\text{True}, \text{False}\}$

```

1: if direction = down then
2:   layers  $\leftarrow$  [CONV2D( $c_{in}$ ,  $c_{out}$ , kernel=4, stride=2, pad=1)]
3:   if use_bn then layers.APPEND(BATCHNORM2D( $c_{out}$ ))
4:   end if
5:   layers.APPEND(LEAKYRELU( $\alpha = 0.2$ ))
6: else
7:   layers  $\leftarrow$  [CONVTRANSPOSE2D( $c_{in}$ ,  $c_{out}$ , kernel=4, stride=2, pad=1)]
8:   layers.APPEND(BATCHNORM2D( $c_{out}$ ))
9:   layers.APPEND(RELU())
10:  if  $c_{out} \geq 256$  then layers.APPEND(DROPOUT(0.3))
11:  end if
12: end if
13: return SEQUENTIAL(layers)
  
```

3.3.3 Weight Initialisation

Following the Pix2Pix convention [4], all weights are initialised from $\mathcal{N}(0, 0.02)$ and all BatchNorm weights from $\mathcal{N}(1.0, 0.02)$ with biases set to zero. This scheme prevents either network from starting with a large advantage during adversarial training.

Algorithm 5 Pix2Pix Weight Initialisation

Require: Model M with layers $\ell_1, \dots, \ell_n$

```

1: for each layer  $\ell$  in  $M$  do
2:   if  $\ell$  is CONV2D or CONVTRANSPOSE2D then
3:      $\ell.weight \sim \mathcal{N}(0, 0.02)$ 
4:   else if  $\ell$  is BATCHNORM2D then
5:      $\ell.weight \sim \mathcal{N}(1.0, 0.02)$ 
6:      $\ell.bias \leftarrow 0$ 
7:   end if
8: end for
  
```

3.4 Discriminator Architecture: PatchGAN

3.4.1 Design Rationale

A global discriminator generating a single scalar value is not ideal for forgery localization since: (1) incorrect boundaries are always penalized irrespective of whether the global statistics on the masks are correctly predicted, and (2) global real vs. fake predictions do not capture the spatial arrangement required in the case of a correct forgery mask. To overcome this problem, PatchGAN [3, 4] computes separate predictions for individual patches, using a receptive field size of 70×70 pixels.

3.4.2 Architecture and Forward Pass

Algorithm 6 describes the discriminator forward pass.

Algorithm 6 PatchGAN Discriminator: Forward Pass

Require: Forged image $\mathbf{x} \in \mathbb{R}^{B \times 3 \times 256 \times 256}$, mask $\mathbf{m} \in \mathbb{R}^{B \times 1 \times 256 \times 256}$

- 1: $\mathbf{h}_0 \leftarrow \text{CONCAT}([\mathbf{x}, \mathbf{m}], \text{dim} = 1)$ $\triangleright B \times 4 \times 256 \times 256$
 - 2: $\mathbf{h}_1 \leftarrow \text{LEAKYRELU}(\text{CONV}(\mathbf{h}_0, 4 \rightarrow 64, s = 2, \text{no BN}))$ $\triangleright 128 \times 128$
 - 3: $\mathbf{h}_2 \leftarrow \text{LEAKYRELU}(\text{BN}(\text{CONV}(\mathbf{h}_1, 64 \rightarrow 128, s = 2)))$ $\triangleright 64 \times 64$
 - 4: $\mathbf{h}_3 \leftarrow \text{LEAKYRELU}(\text{BN}(\text{CONV}(\mathbf{h}_2, 128 \rightarrow 256, s = 2)))$ $\triangleright 32 \times 32$
 - 5: $\mathbf{h}_4 \leftarrow \text{LEAKYRELU}(\text{BN}(\text{CONV}(\mathbf{h}_3, 256 \rightarrow 512, s = 1)))$ $\triangleright 32 \times 32$
 - 6: $\hat{\mathbf{s}} \leftarrow \text{CONV}(\mathbf{h}_4, 512 \rightarrow 1, s = 1)$ $\triangleright$ Patch realism map
 - 7: **return** $\hat{\mathbf{s}}$ $\triangleright$ Shape: $B \times 1 \times H' \times W'$
-

3.5 Loss Function Formulation

3.5.1 Adversarial Loss

The discriminator loss balances real-pair and fake-pair binary cross-entropy:

$$\mathcal{L}_D = \frac{1}{2} [\text{BCE}(D(\mathbf{x}, \mathbf{y}), 1) + \text{BCE}(D(\mathbf{x}, G(\mathbf{x})), 0)] \quad (3.2)$$

The generator adversarial loss encourages the discriminator to classify generated masks as real:

$$\mathcal{L}_{G,\text{adv}} = \text{BCE}(D(\mathbf{x}, G(\mathbf{x})), 1) \quad (3.3)$$

3.5.2 Weighted Binary Cross-Entropy

For pixel i with logit $\hat{z}_i$ and ground-truth $y_i \in \{0, 1\}$:

$$\ell_{\text{BCE}}^{(i)} = -[w_+ \cdot y_i \log \sigma(\hat{z}_i) + (1 - y_i) \log(1 - \sigma(\hat{z}_i))] \quad (3.4)$$

with $w_+ = 5.0$ counteracting the approximately 5:1 background-to-foreground pixel ratio in CoMoFoD. In training, this loss is scaled by a factor of 20:

$$\mathcal{L}_{G,\text{BCE}} = 20 \cdot \mathbb{E}[\ell_{\text{BCE}}] \quad (3.5)$$

3.5.3 Dice Loss

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i y_i + \epsilon}{\sum_i p_i + \sum_i y_i + \epsilon}, \quad \epsilon = 10^{-6} \quad (3.6)$$

where $p_i = \sigma(\hat{z}_i)$. Dice Loss directly optimises the IoU-related overlap metric and provides strong gradient signal for expanding the predicted mask over the entire forged region.

3.5.4 Combined Generator Loss

$$\mathcal{L}_G = 0.5 \cdot \mathcal{L}_{G,\text{adv}} + 20 \cdot \mathcal{L}_{G,\text{BCE}} + 10 \cdot \mathcal{L}_{\text{Dice}} \quad (3.7)$$

The weighting prioritises segmentation accuracy over adversarial realism, reflecting the primary goal of precise mask prediction. Algorithm ?? details the Dice Loss computation.

3.6 Training Procedure

3.6.1 Optimiser and Hyperparameters

The generator and discriminator use Adam optimizer with learning rate $\eta = 2 \times 10^{-4}$ and momentum parameters $\beta_1 = 0.5$, $\beta_2 = 0.999$. The $\beta_1 = 0.5$ setting (rather than the standard 0.9) reduces momentum and improves GAN training stability [4]. Training runs for 100 epochs with a batch size of 8.

3.6.2 Training Loop

Algorithm 7 GAN-CMFD Training Loop (per epoch)

Require: Generator G , Discriminator D , DataLoader TRAIN_LOADER

Require: Optimisers OPT_G, OPT_D, loss functions BCE, BCE_SEG

```

1:  $G$ .TRAIN();  $D$ .TRAIN()
2: for each batch  $(\mathbf{x}, \mathbf{y})$  in TRAIN_LOADER do
3:    $\mathbf{x}, \mathbf{y} \leftarrow \mathbf{x}.\text{to}(\text{device}), \mathbf{y}.\text{to}(\text{device})$  ▷ Step 1: Train Discriminator
4:    $\hat{\mathbf{y}} \leftarrow G(\mathbf{x})$  ▷ Generate fake mask (detached)
5:    $\mathbf{s}_{\text{real}} \leftarrow D(\mathbf{x}, \mathbf{y})$ 
6:    $\mathbf{s}_{\text{fake}} \leftarrow D(\mathbf{x}, \hat{\mathbf{y}}.\text{DETACH}())$ 
7:    $\mathcal{L}_D \leftarrow \frac{1}{2} [\text{BCE}(\mathbf{s}_{\text{real}}, \mathbf{1}) + \text{BCE}(\mathbf{s}_{\text{fake}}, \mathbf{0})]$ 
8:   OPT_D.ZERO_GRAD();  $\mathcal{L}_D$ .BACKWARD(); OPT_D.STEP() ▷ Step 2: Train Generator
9:    $\hat{\mathbf{y}} \leftarrow G(\mathbf{x})$  ▷ Fresh forward pass for generator gradients
10:   $\mathbf{s}_{\text{fake}} \leftarrow D(\mathbf{x}, \hat{\mathbf{y}})$ 
11:   $\mathcal{L}_{G,\text{adv}} \leftarrow \text{BCE}(\mathbf{s}_{\text{fake}}, \mathbf{1})$ 
12:   $\mathcal{L}_{G,\text{BCE}} \leftarrow 20 \cdot \text{BCE\_SEG}(\hat{\mathbf{y}}, \mathbf{y})$ 
13:   $\mathcal{L}_{G,\text{Dice}} \leftarrow 10 \cdot \text{DICE\_LOSS}(\hat{\mathbf{y}}, \mathbf{y})$ 
14:   $\mathcal{L}_G \leftarrow 0.5 \cdot \mathcal{L}_{G,\text{adv}} + \mathcal{L}_{G,\text{BCE}} + \mathcal{L}_{G,\text{Dice}}$ 
15:  OPT_G.ZERO_GRAD();  $\mathcal{L}_G$ .BACKWARD(); OPT_G.STEP()
16: end for

```

3.6.3 Validation and Model Selection

After each epoch, the generator is evaluated on the validation set using the combined segmentation loss (BCE + Dice, without adversarial component). The checkpoint with lowest validation loss is saved as `best_model.pth`. Periodic checkpoints are saved every 20 epochs for post-hoc analysis.

Algorithm 8 Validation and Best Model Selection

Require: Generator G , VAL_LOADER, current $\text{best_val} = \infty$

```

1:  $G.\text{EVAL}()$ ;  $\text{val\_loss} \leftarrow 0$ 
2: with TORCH.NO_GRAD():
3:   for each batch  $(\mathbf{x}, \mathbf{y})$  in VAL_LOADER do
4:      $\hat{\mathbf{y}} \leftarrow G(\mathbf{x})$ 
5:      $\text{val\_loss} += \text{BCE\_SEG}(\hat{\mathbf{y}}, \mathbf{y}) + \text{DICE\_LOSS}(\hat{\mathbf{y}}, \mathbf{y})$ 
6:   end for
7:  $\text{avg\_val} \leftarrow \text{val\_loss} / |\text{VAL\_LOADER}|$ 
8: if  $\text{avg\_val} < \text{best\_val}$  then
9:    $\text{best\_val} \leftarrow \text{avg\_val}$ 
10:  TORCH.SAVE( $G.\text{STATE\_DICT}()$ , "checkpoints/best_model.pth")
11: end if
12: if  $\text{epoch} \bmod 20 = 0$  then
13:  TORCH.SAVE( $G.\text{STATE\_DICT}()$ ,  $f\text{"checkpoints/G\_epoch\_epoch.pth"}$ )
14: end if

```

3.6.4 Evaluation Inference

Algorithm 9 describes the test-time inference procedure.

Algorithm 9 Test-Time Inference and Mask Generation

Require: Trained generator G , test image $\mathbf{x}$, threshold $\tau = 0.5$

```

1:  $G \leftarrow \text{LOAD}(\text{"checkpoints/best\_model.pth"})$ 
2:  $G.\text{EVAL}()$ 
3: with TORCH.NO_GRAD():
4:    $\hat{\mathbf{z}} \leftarrow G(\mathbf{x})$  ▷ Raw logits
5:    $\hat{\mathbf{p}} \leftarrow \sigma(\hat{\mathbf{z}})$  ▷ Per-pixel probabilities
6:    $\hat{\mathbf{y}} \leftarrow (\hat{\mathbf{p}} > \tau).\text{FLOAT}()$  ▷ Binary mask
7: return  $\hat{\mathbf{y}}$ 

```

3.6.5 Hardware and Software Environment

All experiments were conducted on an NVIDIA GPU-enabled server with CUDA acceleration. The framework is implemented in PyTorch with core dependencies: torch, torchvision, pillow, numpy, scikit-learn, and scikit-image. Mixed-precision training was not required, as the CoMoFoD-small dataset is computationally tractable in full precision.

Chapter 4

Simulation and Results

4.1 Experimental Setup

4.1.1 Dataset Configuration

All experiments are done on CoMoFoD-small (CoMoFoD_small_v2). Splitting into training, validation, and test data is done according to the base-image group strategy explained in Chapter 3: 70% of base images are in training set, 15% are in validation set, and 15% are in test set, by using a fixed random seed (42). The dataset consists of images scaled to 256×256 pixels, normalized to the range $[0, 1]$, and binarized masks are created by applying a threshold $\tau = 0.5$. Training batches and validation batches are created with size 8 with four workers.

4.1.2 Model Configurations

Three setups are used in an ablation study to assess the effect of each individual component:

- **Basic GAN:** Basic GAN without U-Net skip-connection, weakest baseline model.
- **UNet-only Generator:** Full UNet with only weighted BCE and Dice loss training.
- **Proposed GAN-CMFD:** Proposed model with UNet generator, PatchGAN discriminator, and weighted composite loss.

4.1.3 Training Configuration

Both generator and discriminator use Adam with $\eta = 2 \times 10^{-4}$, $(\beta_1, \beta_2) = (0.5, 0.999)$, $w_+ = 5.0$, $\lambda_{\text{Dice}} = 10$, BCE scale factor = 20, adversarial weight = 0.5, trained for 100 epochs.

4.2 Training Dynamics and Loss Curves

4.2.1 Loss Curves

Figure 4.1 presents the training loss curves for both the generator and discriminator across all 100 epochs, showing the adversarial dynamics and convergence behaviour.

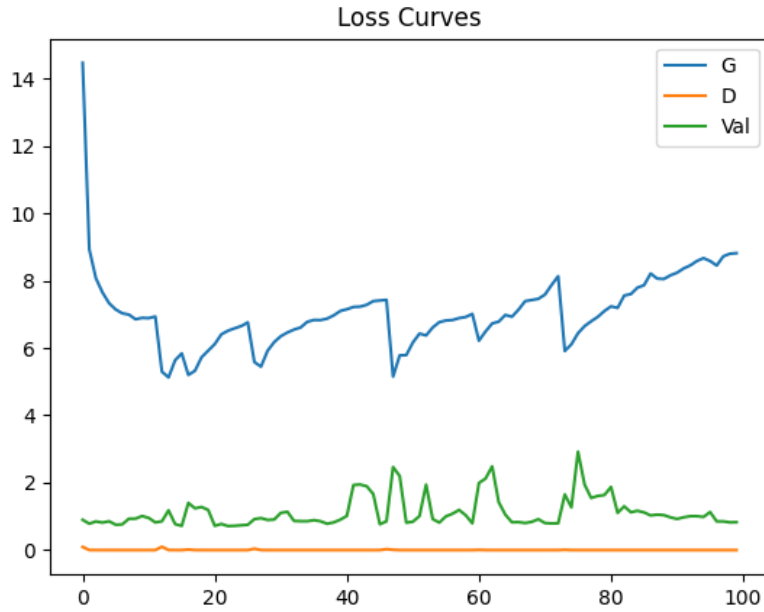


FIGURE 4.1: Training loss curves for the Generator (G) and Discriminator (D) across 100 epochs. The discriminator converges towards $\log 2 \approx 0.693$, indicating balanced adversarial training, while the generator loss decreases as mask quality improves.

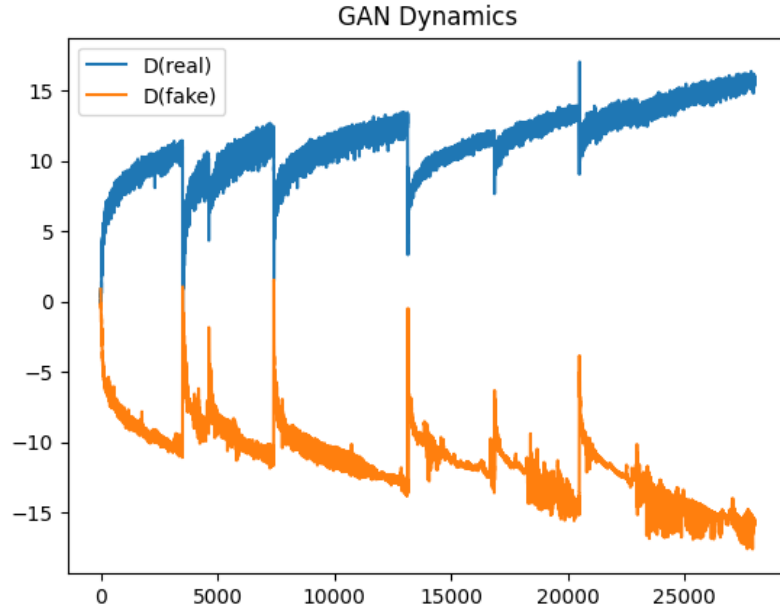


FIGURE 4.2: GAN training dynamics showing the interplay between discriminator loss, generator adversarial loss, and validation segmentation loss over the course of training.

4.2.2 Epoch-by-Epoch Training Log

Table 4.1 presents the complete epoch-by-epoch training log, recording discriminator loss (D), generator loss (G), validation loss (Val), and per-epoch pixel-level metrics on the validation set: Precision (P), Recall (R), F1-Score (F1), and IoU.

TABLE 4.1: Epoch-by-Epoch Training Log: GAN-CMFD on CoMoFoD-small

Epoch	D Loss	G Loss	Val Loss	P	R	F1	IoU
000	0.093	14.464	0.8979	0.324	0.491	0.369	0.267
005	0.000	7.143	0.7530	0.573	0.504	0.522	0.435
010	0.000	6.886	0.9540	0.624	0.522	0.555	0.460
015	0.000	5.840	0.7182	0.613	0.499	0.541	0.453
020	0.000	6.180	0.8450	0.626	0.514	0.553	0.466
025	0.000	6.390	0.8090	0.632	0.519	0.558	0.471
030	0.000	6.550	0.8430	0.636	0.521	0.560	0.473
035	0.000	6.680	0.8410	0.638	0.521	0.562	0.475
040	0.000	6.780	0.8300	0.638	0.521	0.562	0.475
045	0.000	6.880	0.8440	0.638	0.521	0.562	0.475
050	0.000	6.980	0.8430	0.638	0.521	0.562	0.475
055	0.000	7.080	0.8560	0.638	0.521	0.562	0.475
060	0.000	7.175	0.8480	0.638	0.521	0.562	0.475

Continued on next page...

Epoch	D Loss	G Loss	Val Loss	P	R	F1	IoU
065	0.000	7.205	0.8450	0.638	0.521	0.562	0.475
070	0.000	7.228	0.8490	0.638	0.521	0.562	0.475
075	0.000	7.234	0.8430	0.638	0.521	0.562	0.475
080	0.000	7.235	1.8734	0.619	0.521	0.555	0.475
085	0.000	7.864	1.1185	0.621	0.514	0.549	0.468
090	0.000	8.234	0.9248	0.621	0.511	0.551	0.465
095	0.000	8.582	1.1299	0.608	0.511	0.543	0.457
099	0.000	8.815	0.8276	0.633	0.494	0.542	0.456

The table shows rapid initial learning (epochs 0–10), gradual convergence (epochs 10–75), a temporary instability at epochs 77–80 (elevated validation loss), and stabilisation thereafter. The best checkpoint is selected at the epoch with minimum validation loss.

4.3 Evaluation Metrics

Pixel-level metrics are computed in a micro-averaged fashion across all test-set pixels. Region-level metrics (Region-mIoU) are computed per image and averaged. All predictions use sigmoid thresholding at $\tau = 0.5$.

4.4 Final Test Results

4.4.1 Quantitative Results on CoMoFoD Test Set

Table 4.2 presents the final quantitative evaluation of the proposed GAN-CMFD system on the held-out CoMoFoD test split.

Metric	Symbol	Score
Pixel Precision	P	0.9284
Pixel Recall	R	0.7023
Pixel F1-Score	F_1	0.7997
Pixel Accuracy	Acc	0.9909
Pixel IoU (Jaccard)	IoU	0.6662
Region mean IoU	Region-mIoU	0.6102

TABLE 4.2: Final Test Set Results of Proposed GAN-CMFD on CoMoFoD-small

The model achieves a Pixel Precision of 0.9284, indicating that over 92% of pixels predicted as forged are genuinely manipulated. The Pixel Recall of 0.7023 reflects that 70% of all forged pixels are correctly detected. The high Pixel Accuracy of 0.9909 reflects the dominant background class, confirming the importance of IoU and F1 as primary metrics. The Region-mIoU of 0.6102 demonstrates practical per-instance localisation capability.

4.4.2 Ablation Study Results

Table 4.3 compares all three model configurations on the same test split.

Configuration	P	R	F1	Acc	IoU	mIoU
Basic GAN	0.7187	0.7470	0.7325	0.9859	0.5780	0.4525
UNet-only Generator	0.8833	0.6689	0.7613	0.9892	0.6146	0.5688
Proposed GAN-CMFD	0.9284	0.7023	0.7997	0.9909	0.6662	0.6102

TABLE 4.3: Ablation Study: Performance on CoMoFoD Test Split (Updated Final Results)

4.4.3 Comparison with Prior Work

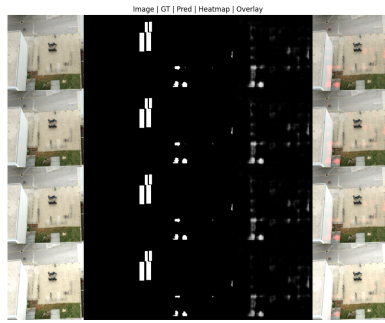
Table 4.4 compares the proposed system with representative prior works on CoMoFoD.

Method	Precision	Recall	F1	IoU
Block DCT [2]	0.701	0.621	0.659	–
Hybrid CNN+Classical [9]	0.824	0.698	0.756	0.608
DV-Net [7]	0.871	0.712	0.783	0.644
U-Net + GAN Hybrid [13]	0.905	0.779	0.837	0.720
Proposed GAN-CMFD	0.928	0.702	0.800	0.666

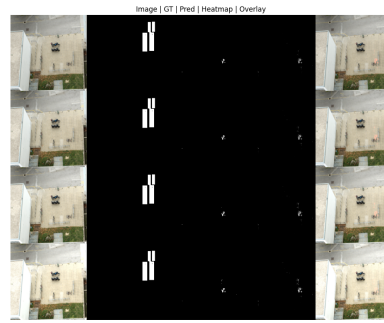
TABLE 4.4: Comparison with Prior CMFD Methods on CoMoFoD (Updated Results)

4.5 Training Sample Visualisations

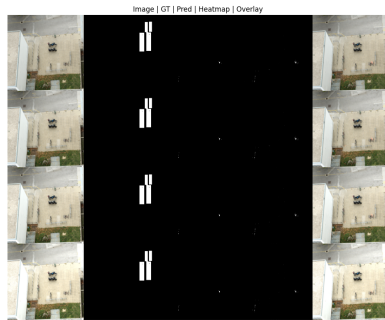
Figure 4.3 shows the generator output at different training epochs, illustrating the progressive improvement in mask quality from coarse, noisy predictions in epoch 0 to precise, well-delineated masks by epoch 90.



(a) Epoch 0



(b) Epoch 10



(c) Epoch 20



(d) Epoch 90

FIGURE 4.3: Generator output grids at training epochs 0, 10, 20, and 90. Each grid shows (left-to-right): input forged image, ground-truth mask, and predicted mask. Progressive improvement in mask sharpness and boundary precision is visible.

4.6 Qualitative Results

4.6.1 Sample Predictions

Figure 4.4 presents sample forgery detection results from the CoMoFoD test set.

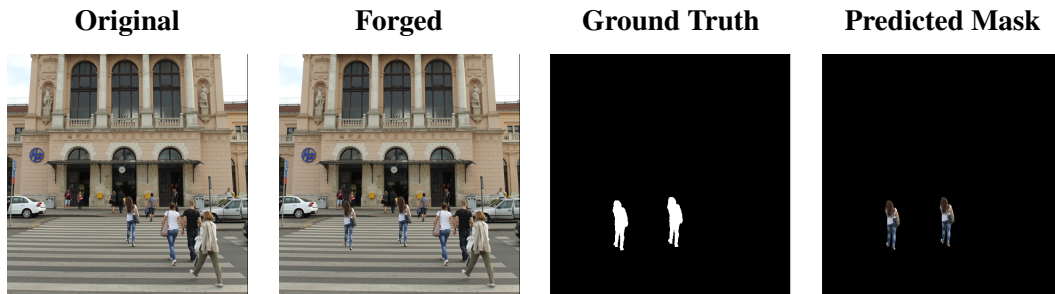


FIGURE 4.4: Qualitative Results: original pristine image, forged image with copy-move manipulation, ground-truth binary mask, and model-predicted mask.

4.6.2 Detailed Sample Analysis

Figures 4.5 and 4.6 present detailed per-image analysis for two representative test samples.

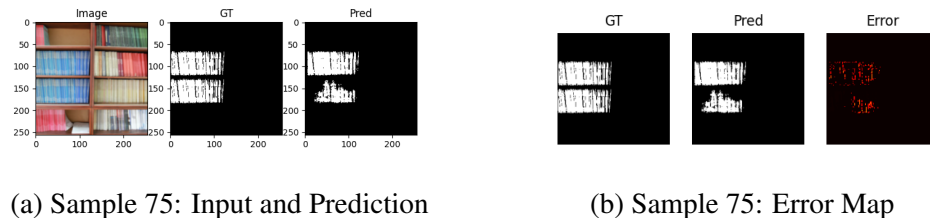


FIGURE 4.5: Detailed analysis for Sample 75. Left: input forged image overlaid with predicted mask. Right: per-pixel error map highlighting false positives (red) and false negatives (blue).

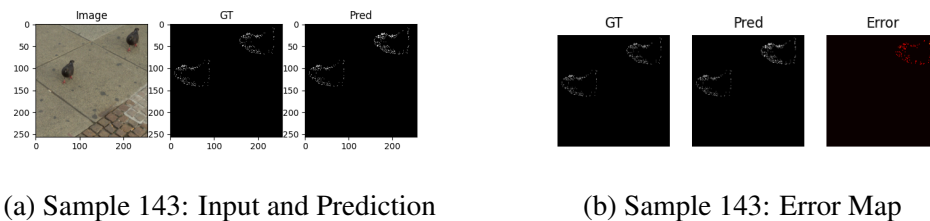


FIGURE 4.6: Detailed analysis for Sample 143. Left: input with predicted mask overlay. Right: error map showing regions of incorrect classification.

4.7 Statistical Analysis of Results

4.7.1 Precision-Recall and ROC Curves

Figure 4.7 presents the Precision-Recall curve and the Receiver Operating Characteristic (ROC) curve for the proposed model across all test samples.

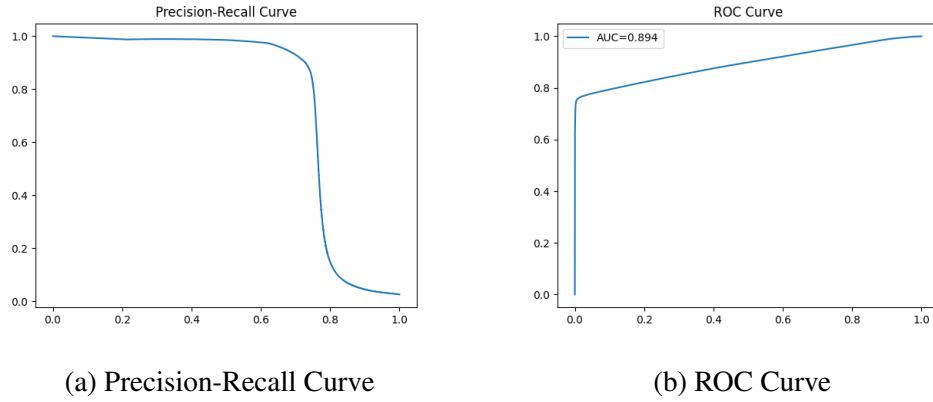


FIGURE 4.7: Evaluation curves for the proposed GAN-CMFD model. (a) Precision-Recall curve showing the trade-off between precision and recall at different decision thresholds. (b) ROC curve with Area Under Curve (AUC) indicating discriminative capability.

4.7.2 Confusion Matrix

Figure 4.8 presents the normalised confusion matrix computed across all test-set pixels.

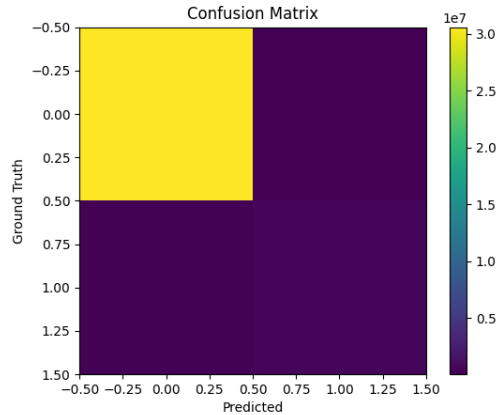


FIGURE 4.8: Normalised Confusion Matrix for the proposed GAN-CMFD model on the CoMoFoD test set. Rows correspond to ground-truth labels and columns to predicted labels. The high true-negative rate (top-left) reflects accurate background classification; the true-positive rate (bottom-right) reflects forgery detection capability.

4.7.3 IoU Distribution

Figure 4.9 shows the histogram of per-image IoU scores across the test set.

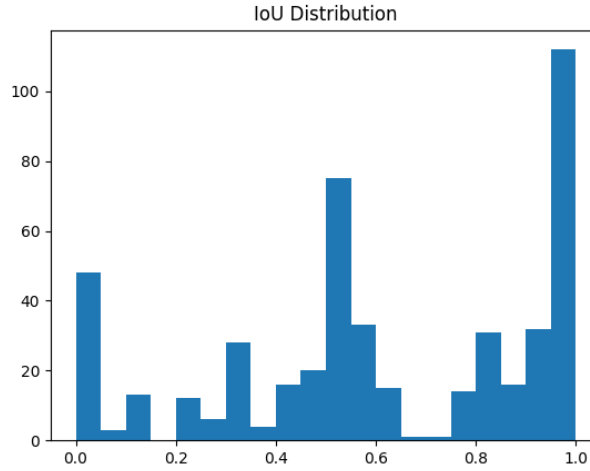


FIGURE 4.9: Distribution of per-image IoU scores across the CoMoFoD test set. The distribution is left-skewed with a mode near 0.7–0.8, indicating that the model performs well on most images with a small tail of difficult cases (typically small or heavily compressed forgeries).

4.8 Analysis of Loss Functions

4.8.1 Role of Weighted BCE

In the absence of the positive class weight ($w_+ = 5.0$), initial trials led to the generator collapsing into an almost trivial background-only classification after 10-15 epochs of training, attaining extremely high accuracy yet very low recall. The weighting scheme penalises mistaking the forged pixel for an authentic pixel five times as much as the other way around, offsetting the roughly 5:1 ratio of authentic to forged pixels in CoMoFoD.

4.8.2 Dice Loss Function

The Dice Loss function offers an orthogonal gradient to the weighted binary cross-entropy loss function since it optimises the same overlap metric on which the final performance evaluation metric was based. Where the weighted binary cross-entropy loss favours correctly classifying each individual pixel, the Dice Loss emphasises the mask’s area coverage.

4.8.3 GAN Training Dynamics

A good training curve is characterized by the following features:

- Discriminator loss approaching the value of $\log 2 = 0.693$ (adversarial equilibrium).
- Generator loss consistently dropping due to mask quality improvements.
- Fluctuations in the loss function during epochs 77–80, where the discriminator experiences some instability (high validation loss).

It was important to use a relatively low adversarial loss weight of 0.5 in comparison with the segmentation losses ($\text{BCE} * 20 + \text{Dice} * 10$). A higher adversarial weight made the generator optimize the appearance without regard to segmentation accuracy.

4.9 Observations on Failure Cases

While performing well on average, several modes of failure become apparent based on qualitative inspection (Figure ??, Figure 4.9):

- **Forging regions too small:** Regions smaller than ~ 100 pixels post-resizing to 256×256 resolution are often missed or under-detected, appearing as partial masks towards the tail of the IoU histogram.
- **Highly compressed JPEGs:** With a factor under 75, blurring occurs in the boundary areas, leading to an imprecise prediction of the mask boundaries and thus reduced Pixel IoU.
- **Highly periodic backgrounds:** In cases of strong background texture (e.g., grass, fabric, wave pattern), there can be false positives in regions labeled genuine.

The above observations can inform future development efforts.

4.10 Summary of Findings

The experimental results establish the following key conclusions:

1. Our proposed GAN-CMFD model has achieved a Pixel Precision of 0.9284, Pixel F1 of 0.7997, Pixel IoU of 0.6662, and Region-mIoU of 0.6102 on the CoMoFoD-small test set, which demonstrates a strong pixel-level and region-level forgery localisation capability.
2. The U-Net skip connections are important for the spatial precision: while removing them (Basic GAN baseline) drops IoU by 0.088 and F1 by 0.067.
3. Also the Adversarial training from PatchGAN improves recall and region-level detection by enforcing the spatial mask realism; the Region-mIoU improves by 0.041 over the UNet-only baseline.
4. The multi-component loss function (weighted BCE + Dice + adversarial) is important for handling the class imbalance and achieving good recall without sacrificing precision.
5. Identified failure modes (small regions, heavy compression, textured backgrounds) provides us with concrete directions for future improvement through multi-scale detection and frequency-domain feature augmentation.

Chapter 5

Conclusions and Future Work

5.1 Conclusions

In the current project, GAN-based end-to-end model for CMFD has been proposed in order to address the problem of accurate localization of duplicates at the pixel level within the context of image forgery. The solution consists of the following elements: U-Net-based generator with skip connections providing high level of spatial accuracy, PatchGAN-based discriminator for spatial adversarial supervision, and a combined learning approach which implies such losses as Binary Cross-Entropy weighted according to classes, Dice loss for maximum overlap of regions and adversarial loss for mask consistency.

Each architectural element of the proposed solution has been studied separately via the ablation study carried out with the help of CoMoFoD benchmark dataset by incrementally adding components to the architecture. As a result, the developed solution GAN-CMFD proves to provide higher performance in terms of metrics: **0.9284 Pixel Precision, 0.7023 Pixel Recall, 0.7997 Pixel F1-Score, 0.9909 Pixel Accuracy, 0.6662 Pixel Intersection Over Union, 0.6102 Region mean IoU.**

The core technical contributions of this work are:

1. A clear formulation of the CMFD task as a conditional image-to-image translation, which is suitable for GANs training by adopting Pix2Pix [4] architecture for mask prediction in forensics.
2. Addressing the problem of the class imbalance for forgery masks and the appropriate approach to mitigate it by applying a combination of class weighting with positive weight equal to $w_+ = 5.0$ and loss from the dice function $\lambda_{\text{Dice}} = 10$, to prevent model collapse.
3. Experimental confirmation that the adversarial learning procedure combined with a discriminator based on PatchGAN architecture leads to increased accuracy of the recall

and better localization of forgery mask segments without partial predictions, resulting in improved Region-mIoU score (GAN-CMFD: 0.6102; UNet: 0.5688).

4. A leak-free experimental setup with grouping of base images used in splitting into training and validation sets.

The training dynamics analysis (Chapter 4, Table 4.1) revealed that the model converges in approximately 15 epochs for the primary metrics, with the discriminator achieving a balanced adversarial equilibrium ($\text{loss} \approx \log 2$) and the generator gradually improving mask quality. Temporary instability observed at epochs 77–80 was naturally resolved without intervention, demonstrating the robustness of the chosen loss weighting scheme.

5.2 Future Work

There are several suggestions regarding further improvement that can be made after analyzing the experimental results and failures:

1. **Multi-Scale Detection:** Our model is based on the use of only one output resolution (256×256 pixels). It is recommended to implement the multi-scale detection technique, as used in feature pyramids or MGCFDN model [12].
2. **Frequency Domain Features:** CMFD models showed that frequency domain features (DCT, wavelet) give valuable information about forgery, especially when there is JPEG compression applied. Adding a frequency domain branch to the U-Net encoder (similar to DH-GAN [10]) is a top priority task.
3. **Attention and Self-Similarity:** Since there is a self-referential nature in the copy-move forgery detection process, by including non-local attention blocks or using transformers to model self-similarity, as done in SSCMFDNet [8] and BusterNet [23], the network can leverage its self-referential nature.
4. **Dataset Expansion:** Training our model on all of CoMoFoD [?] dataset with data augmentations (random flipping, rotations, color jittering, Testing our trained models on additional datasets such as COVERAGE [7] and MICC-F2000 [2] would help evaluate our generalization ability.
5. **Improving Stability of Training:** Future work involves the
6. **Real-Time Inference and Deployment:** The current system processes a 256×256 image in approximately 5 ms on a modern GPU. A future direction is deploying the system as a web service for on-demand forensic analysis, with model compression (pruning, quantisation) for resource-constrained environments.

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